

Facility Operations & Safety Management

STANDARD OPERATING PROCEDURE (SOP)

Purpose

This Standard Operating Procedure (SOP) defines the mandatory process for managing daily building operations, ensuring staff compliance, and responding to predictive system alerts across all facility zones. The objective is to maintain a safe, sterile, and fully operational environment—particularly within critical areas such as operating rooms—while preventing asset failure and ensuring staff safety.

Scope and Applicability

This SOP applies to the entire facility, encompassing general public areas, staff-only zones, mechanical rooms, and critical sterile environments (e.g., Operating Rooms). It must be followed by all facility management, operational staff, and maintenance personnel. Interventions may be routine or triggered by the AI Engine and delivered via the M365 Copilot interface.

Roles and Responsibilities

- **Facilities Manager** - Responsible for overall building compliance, approving work order closures, and ensuring safety standards are met across all departments.
- **Maintenance Technician** - Responsible for executing facility checks, responding to AI-dispatched work orders, and maintaining building systems (HVAC, plumbing, electrical).
- **Clinical/Operations Lead** - Responsible for ensuring operating rooms and staff areas meet specific environmental and cleanliness standards prior to daily use.
- **Health & Safety Officer** - Responsible for auditing staff safety compliance and ensuring emergency routes and equipment are unobstructed.

Overview

This SOP outlines the standard daily process to inspect, monitor, and document the holistic health of the building. It ensures that standard environments and critical operating rooms are maintained within strict parameters, staff are adhering to safety protocols, and any AI-generated predictive maintenance alerts are addressed promptly.

Trigger Conditions (Entry Criteria)

This SOP is executed daily as a routine procedure, but immediate interventions are initiated when the following conditions are met:

- Building Management System (BMS) detects environmental anomalies (e.g., improper positive pressure or humidity in operating rooms).
- A life-safety system or security access point registers a fault.
- A predictive maintenance alert (Smart Action Pack) is issued by the AI Engine regarding general building infrastructure.

Health & Safety Rules

- Standard site Personal Protective Equipment (PPE) must be worn in designated mechanical and active maintenance zones.
- Strict aseptic gowning and sterilization protocols must be followed before entering any operating room or critical clinical zone.
- All fire exits, emergency evacuation routes, and spill containment kits must be physically verified as unobstructed daily.
- Any hazardous material spills or biological safety breaches must immediately halt work and trigger standard site safety isolation procedures.

Required Tools

- Mobile device with M365 Copilot interface and D365 auditing application.
- Environmental sensor calibration tool (for verifying OR humidity/pressure).
- Standard site PPE and sterile cleanroom garment kits.
- Master facility access keys/fobs.

Step-by-Step Execution Process

- **1. Preparation:** Review the daily facility health summary and any active predictive maintenance tickets generated by the AI Engine on your mobile device.
- **2. General Safety Sweep:** Walk through all common areas and staff zones. Verify that all fire exits are clear, emergency lighting is functional, and fire extinguishers are present and pressurized.
- **3. Critical Zone (Operating Rooms) Inspection:** Enter the operating suite anteroom. Verify that positive pressure indicators are functioning. Check that HVAC temperature and humidity levels are strictly within clinical parameters.

- **4. Staff & Security Compliance Check:** Ensure all staff in restricted areas are displaying proper identification and adhering to zone-specific PPE or sterile dress codes. Verify secure doors are latching correctly.
- **5. System Interventions:** If a specific alert was triggered (e.g., fluctuating water pressure or abnormal electrical draw in a wing), navigate to the asset and isolate the issue following asset-specific LOTO procedures.
- **6. Resolution & System Restart:** Once any corrective maintenance is complete, clear the work area, restore power, and monitor the local environment to ensure stabilization.

Exit Criteria

This SOP may be considered complete only when all of the following conditions are met:

- All operating rooms and critical zones are verified to be within strict temperature, humidity, and pressure parameters.
- General safety routes are confirmed clear and operational.
- Any triggered predictive maintenance tasks have been resolved and systems have normalized.
- All checklist items are completed and logged.

Exception Handling and Escalation

If critical environments (like ORs) fail to stabilize, or a major safety violation is found:

- Immediately lock down the affected operating room or zone. Do not allow clinical procedures to begin.
- Escalate immediately to the Facilities Manager and the Clinical/Operations Lead.
- Document the persistent fault and initiate emergency repair or hazard containment workflows.

Post-Job Audit and Verification

All daily findings, safety verifications, and corrective actions must be logged in the D365 work order record. Checklist items for critical zones (like OR pressure validation) must be explicitly signed off within the M365 Copilot interface. The daily log must not be closed until all evidence is verified and approved by the Site Supervisor.

SOP Governance

SOP Owner: Facilities Management Team

Version: 1.0

Approval Authority: Director of Operations

Effective Date: 05/05/2026

Review Cycle: Bi-Annual or upon major facility upgrade